

João de Campos Carvalho

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SUMMARY OF SKILLS

Python, C#, C++, TypeScript, Rust, MATLAB, Fusion 360, SolidWorks, PyTorch, EasyEDA, Git, Linux, Figma, Power BI, Microsoft Office

EDUCATION

University of Toronto, BSc, Mechanical Engineering + PEY Co-op Sep 2024 – Apr 2029
Minors in Robotics & Mechatronics and Engineering Business. 3.48 GPA

RELEVANT PROFESSIONAL EXPERIENCE

Auditorium AI Inc., Toronto, Canada

May 2026 – August 2026

Lead Design Engineer

- Engineered a spatial-calibration feature mapping automated stage lighting to room geometry, uncovering two silent failure modes that every existing check had passed.
- Designed a unified interface design system covering typography, components, colour surfaces and a custom icon set, adopted across the operator application and marketing site.
- Conducted independent market and competitive research that redirected the company's target customer segment, and produced the follow-up channel strategy.
- Authored an eight-module training curriculum for operators new to lighting consoles.
- Integrated hardware and software for a single-board-computer field device, including field updates, device driver rules and a 3D-printed enclosure designed in Fusion 360.

UTESCA, Toronto, Canada

September 2025 – current

Director of Outreach

- Identified potential partners at prestigious consulting firms, with an 80% conversion rate.
- Developed an automated process in a Google Sheets database to optimize partner outreach.

PROJECT EXPERIENCE

Materials Science Summer Research Project - Co-Author

May – July 2025

- Researched over 20 applications of 4D printing for Li-Ion batteries and supercapacitors, compiling data from 350+ sources.
- Awarded the People's Choice Award at U of T Undergraduate Engineering Research Day.

UHN Sleep Monitoring Device - Co-Project Manager

September – December 2024

- Coordinated a 5-person team designing a low-cost sleep monitoring device with UHN.
- Designed the CAD model in Fusion 360 and integrated the electronic components.
- Originated 120+ design ideas, resulting in a design 30% cheaper than industry competitors.

EXTRACURRICULARS

Brazilian Robotics Olympics - Team Leader / Chief Engineer

2021 – 2023

- Designed and programmed the C# code for the simulation category, placing top 5 nationally.
- Designed and manufactured the robot's custom PCB, circuitry and rescue vision module.

Awards: Bronze medalist - Brazilian Mathematics Olympiad | Silver medalist - Mathematique San Frontieres